Department of Electronics and Communication Engineering
Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: VII Semester B.E ECE A
Course Code & Title: EC8702 Adhoc and Wireless Sensor Networks
Name of the Faculty member (s): Mr.P.Gunasekaran

Innovative Practice Description

- **Unit / Topic: Unit IV / Network Security Attacks**
- **Course Outcome:** CO 4
- **Topic Learning Outcome:** TLO 11
- **Activity Chosen: Flipped Classroom**
- **Justification:**

Network attacks are unauthorized actions on the digital assets within an organizational network. Malicious parties usually execute network attacks to alter, destroy, or steal private data. Perpetrators in network attacks tend to target network perimeters to gain access to internal systems. There are two main types of network attacks: passive and active. In passive network attacks, malicious parties gain unauthorized access to networks, monitor, and steal private data without making any alterations. Active network attacks involve modifying, encrypting, or damaging data.

A sequence of evaluations is used in a collaborative learning strategy to help pupils master a subject or topic. It focuses on successful group interaction among students in order for them to grasp certain subjects. Additionally, this teaching method improves students' communication and problem-solving abilities. Because the textbook only has five pages, it does not receive additional attention in parts of the syllabus and test. As a result, I consider this a matter for independent research. Because all students value independence and want to learn on their own.

- **Time Allotted for the Activity:** 50 Minutes
- **Details of the Implementation:**

A six-person heterogeneous group was created. There were clever pupils, ordinary students, and slow learners in each group. Academic performance determines whether a pupil is classified as a brilliant student, an average student, or a slow learner. The total number of students in the class is 36. (Boys and Girls).

Specific topic was given to the specific group to learn on their own. Resources like reference book, videos were given to the student group. Students are asked to prepare more in depth than before. Get the class back together to share the individual group’s work with everyone.

Sub Topics:

- Introduction to Network Security Attacks
- Physical Layer Attacks
- Link Layer Attacks
- Network Layer attacks
- Transport layer Attacks
- Application Layer Attacks

- **CO – PO / PSO mapping:**

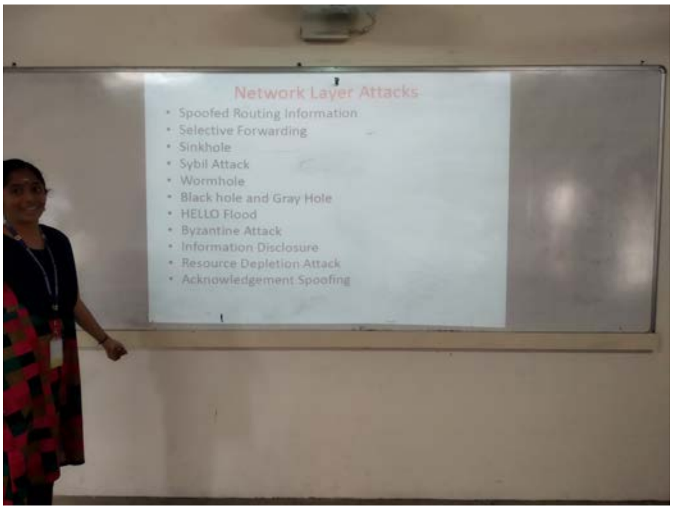
CO	PO1	PO2	PO3	PSO1	PSO2	PSO3
CO2	**3**	**2**	**2**	**3**	**2**	**1**

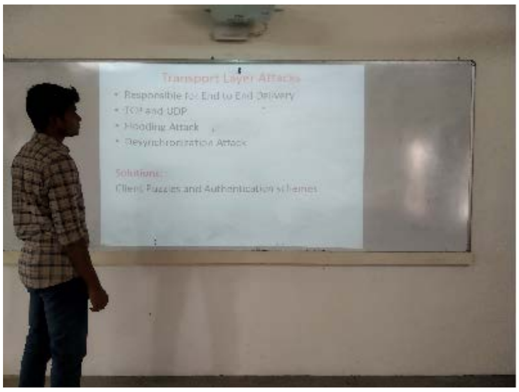
(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO1**	**PSO1**
	3	**3**
Justification for correlation	The fundamentals of cryptography and security is required to understand the concepts of network security attacks	To provide solutions to various network security attacks the fundamentals of cryptography and security is required

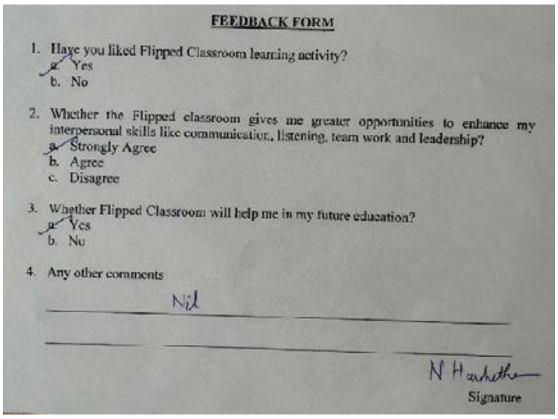
- **Images / Screenshot of the practice:**





- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***

The feedback was collected from the students using the following questionnaires. It is used to assess the success of the Flipped Classroom. The feedback was collected from the students using the following questionnaires. It is used to assess the success of the Flipped Classroom. The first question is used to assess the student's willingness whether the students are like this activity or not. The second and third questions measure the objectives and outcomes of this activity. The fourth question asks the students comments to improve the learning strategies.



FEEDBACK FORM

1. Have you liked Flipped Classroom learning activity?
   a. Yes ✓
   b. No
2. Whether the Flipped classroom gives me greater opportunities to enhance my interpersonal skills like communication, listening, team work and leadership?
   a. Strongly Agree ✓
   b. Agree
   c. Disagree
3. Whether Flipped Classroom will help me in my future education?
   a. Yes ✓
   b. No
4. Any other comments

Nil

N Harshitha

Signature

  - ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

    Every student had an equal opportunity to participate in this activity. The activity's success was assessed by asking the same question in Internal Assessment Test II – Around 85% of students correctly answered.

  - ***Challenges faced in implementation:***

    The fundamental issue is that few students have been exposed to flipped classrooms. Students struggle with self-discipline and may show up to class without having fully comprehended the material. To encourage all students to participate, I began by praising others, taking turns for equitable involvement, and allowing students to make decisions together.

References:

- ❖ Feng Zhao, Leonidas Guibas, ―Wireless Sensor Networks: an information processing approach, Elsevier publication, 2004.
- ❖ I.F. Akyildiz, W. Su, Sankarasubramaniam, E. Cayirci, ―Wireless sensor networks: a survey, computer networks, Elsevier, 2002, 394 - 422.
- ❖ Stephanie butler velegol, sarah e. Zappe, emily mahoney, The Evolution of a Flipped Classroom: Evidence-Based Recommendations, winter 2015, pp.1-37.
- ❖ Flipped Learning Network (FLN). (2014) The Four Pillars of F-L-I-P™

Signature of Faculty Member **HOD**

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Course Code & Title: EC8702 Adhoc and Wireless Sensor Networks
Name of the Faculty member (s): Mr.P.Gunasekaran

Innovative Practice Description

- **Unit / Topic: Unit V / Sensor Node Hardware**
- **Course Outcome:** CO 5
- **Topic Learning Outcome:** TLO 13
- **Activity Chosen: Three Step Interview**
- **Justification:**
  Sensor node hardware can be grouped into three categories, each of which entails a different set of trade-offs in the design choices.
  - Augmented general-purpose computers
  - Dedicated embedded sensor nodes
  - System-on-chip (SoC) nodes

  Three-Step Interview has student pairs take turns interviewing each other, then asks them to report what they learned to another pair. It ignites the students to improve listening, writing, and thinking skills by questioning each other, sharing their ideas, and writing notes.
- **Time Allotted for the Activity:** 50 Minutes
- **Details of the Implementation:**
  - Divide students into groups of three include a bright, one average student, and one slow learner.
  - Assign three roles: interviewer, interviewee, and note-taker.
  - Assign 10 minutes for Think time for the course content
  - After assigning a theme of discussion, have students participate in a 10-minute interview to discuss what they found to be the key information relating to the topic.
  - After each interview, pair switches roles. Students were grouped based on their grade level (5 minutes).
  - Pairs pair to form a group of six each student in turn, shares with the teammate what he/she learned interview in another 10 minutes.
  - Two students from each group were called and present their ideas about what they discussed and learned. (15 minutes).

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO5	PSO1	PSO2	PSO2
CO2	2	1	2	2	3	3	1

(1 – Low 2 – Moderate 3 – High)

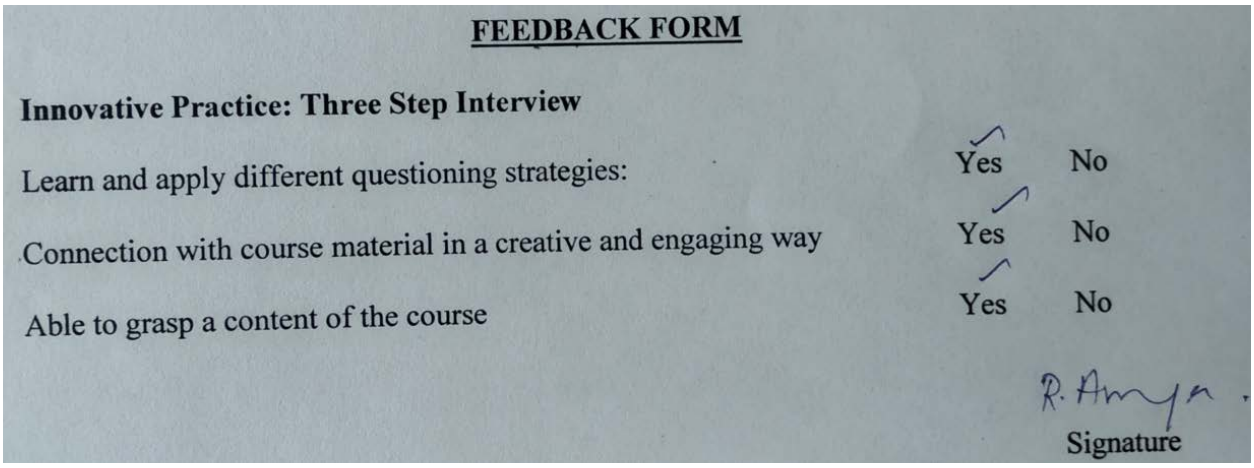
- **PO / PSO mapped:**

Innovative practice	PO5	PSO2
	2	3
Justification for correlation	The modern tools like Energia/CCS are required to create efficient sensor node hardware with effective utilization of resources.	Analysis and Design Procedures of sensor node hardware helps to develop tiny and efficient sensor nodes for real time applications.

- **Images / Screenshot of the practice:**

- **Reflective Critique:**
  - ❖ ***Feedback of practice from students and other stakeholders:***

    From the feedback analysis, students Learn and apply different questioning strategies and Able to grasp a content of the course.



FEEDBACK FORM

Innovative Practice: Three Step Interview

Learn and apply different questioning strategies:	Yes ✓	No
Connection with course material in a creative and engaging way	Yes ✓	No
Able to grasp a content of the course	Yes ✓	No

R. Amya

Signature

- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
  - It encourages students to think about the information they have been taught.
  - It expands their ability to question, think and generate answers.
  - It provides each student with the opportunity to voice their opinions, and promotes equal participation.
  - It promotes student accountability
- ❖ ***Challenges faced in implementation:***
  - To make slow learner to involve the activity effectively.
  - Noise and flexible arrangement

References:

- ❖ Feng Zhao, Leonidas Guibas, ―Wireless Sensor Networks: an information processing approach, Elsevier publication, 2004.
- ❖ I.F. Akyildiz, W. Su, Sankarasubramaniam, E. Cayirci, ―Wireless sensor networks: a survey, computer networks, Elsevier, 2002, 394 - 422.
- ❖ https://kb.wisc.edu/instructional-resources/search.php?cat=11166

Signature of Faculty Member **HOD**

RAMCO INSTITUTE OF TECHNOLOGY

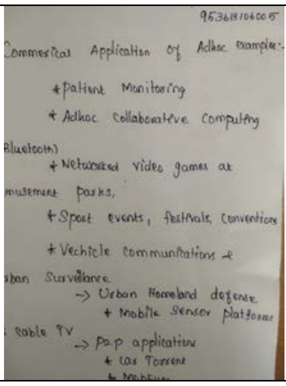
Department of Electronics and Communication Engineering

Academic Year: 2021- 2022 (Odd Semester)

INNOVATIVE TEACHING METHOD

Degree, Semester& Branch: VII Semester B.E.ECE A
Course Code & Title: EC8702 Adhoc and Wireless Sensor networks
Name of the Faculty member: Mr.P.Gunasekaran

Innovative Teaching Method Execution
UNIT I
Elements of Ad hoc Wireless Networks – Class Poll on 18.08.2021

Example commercial applications of Ad hoc networking Exit Slip on 24.08.2021
 953618106005 Commerical Application of Adhoc examples:- * Patient Monitoring * Adhoc Collaborative Computing Bluetooth) * Networked video games at musement parks. * Sport events, festivals, conventions * Vechicle communications & Urban Survillance -> Urban Homeland defense * Mobile Sensor platforms cable TV -> P2P applications * Car Torrent
AODV – Zero Minute Speech on 07.09.2021


Prepared by: Mr.P.Gunasekaran
Date:

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Date:

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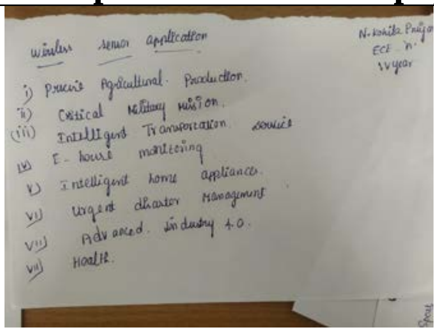
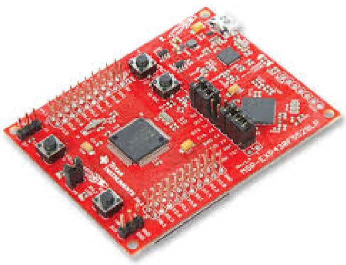
Innovative Teaching Method Execution

UNIT II

WSN application examples – One Minute paper on 14.09.2021



Single-Node Architecture – Hardware Demonstration on 16.09.2021



Optimization Goals and Figures of Merit – Quiz Dominos on 23.09.2021

Prepared by: Mr.P.Gunasekaran
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Date:

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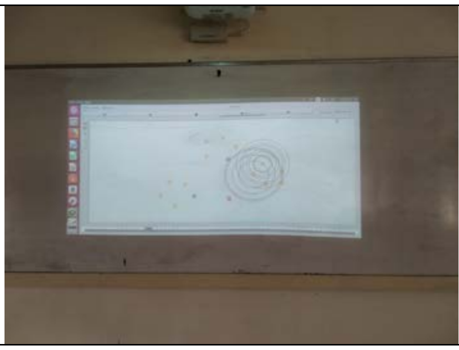
Innovative Teaching Method Execution

UNIT III

MAC Protocols for Wireless Sensor Networks – Quescussion on 28.09.2021

Schedule based protocols – LEACH – Simulation on 08.10.2021



Challenges and Issues in Transport layer protocol – Animated Video on 13.10.2021

Prepared by: Mr.P.Gunasekaran
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Approved by: Dr.S.Periyanayagi
Date:

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Innovative Teaching Method Execution
UNIT IV
Layer wise attacks in wireless sensor networks – Mind Map on 26.10.2021

Black hole attack, flooding attack – Role Play on 02.11.2021

Key Distribution and Management – Think-Pair-Share on 09.11.2021


Prepared by: Mr.P.Gunasekaran
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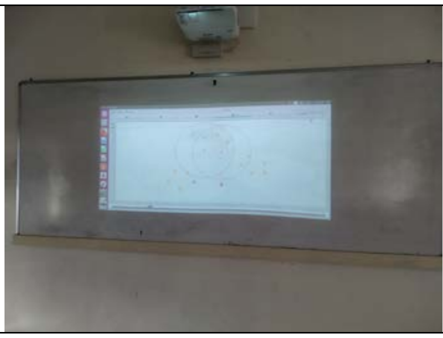
Innovative Teaching Method Execution
UNIT V
Sensor Node Hardware – JAM on 11.11.2021

Node-level software platforms – TinyOS – Pro-Con- Grid on 16.11.2021

NS2 and its extension to sensor networks – Simulation on 19.11.2021


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